

CSE331 Assignment 02 Summer 2024: Regular Expression

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Q1. Which of the following statements about regular expressions is true?

2 points

- ☐ Regular expressions can describe only Regular Languages.
- ☐ Regular expressions can describe only Context Free Languages.
- ☐ Regular expressions can describe both Regular Language and Context Free Languages.

Write an explanation for your answer.

Your answer

Q2. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^* : 2 \text{ points } 0^n, \text{ where } n \text{ is even}\}$?

- ☐ $0(00)^*$
- ☐ $(00)^*$
- ☐ $0^*(00)^*0^*$
- ☐ None of the above

Write an explanation for your answer.

Your answer

Q3. Which is the correct regular expression for the language $L = \{w \in \{0,1\}^* : 2 \text{ points } 0^n1^n, \text{ where } n \text{ is even}\}$?

- ☐ $(00)^*(11)^*$
- ☐ $((0 \cup 1)(0 \cup 1))^*$

☐ $(00 \cup 11)^*$

☐ None of the above

Write an explanation for your answer.

Your answer _____

Q4. You are given two languages, A and B. You can construct a regular expression for A. However, it is impossible to construct a regular expression for B. Now read the options below.

2 points

- a) A is a regular language
- b) B is a regular language.
- c) B is a nonregular language.
- d) $A \cup B$ is a regular language.

Choose the best answer from below.

- ☐ a and b
- ☐ a and c
- ☐ a and d
- ☐ a, b and d
- ☐ a, c and d

Write an explanation for your answer.

Your answer _____

Q5. What does the regular expression $(0 \cup 1)^* 0 (0 \cup 1)^*$ describe?

2 points

- ☐ $L = \{w \in \{0,1\}^*: w \text{ contains exactly one } 0\}$
- ☐ $L = \{w \in \{0,1\}^*: w \text{ contains at least three } 0\}$
- ☐ $L = \{w \in \{0,1\}^*: w \text{ contains at least one } 0\}$
- ☐ $L = \{w \in \{0,1\}^*: w \text{ contains at most one } 0\}$

Write an explanation for your answer.

Your answer

Q6. $(0 \cup 1)^* (0 (0 \cup 1) 0 \cup 1 (100 \cup 001 \cup 11^*)) 00$

2 points

Write down the shortest nonempty string generated by the given expression. Write down the string only.

Your answer

Write an explanation for your answer.

Your answer

Q7. $(00 \cup 01) (101)^* ((00)^* \cup 110)$

2 points

What is the shortest string generated by the regular expression with exactly three 0s? Write down the string.

Your answer

Write an explanation for your answer.

Your answer

Q8. Does the string **00100001011** belong to the regular expression **$0(0 \cup 1)^* 01 (0 \cup 1)^* ((0 \cup 1)^*)^* 11 (0 \cup 1)^* 1$** ?

2 points

☐ Yes

☐ No

Write an explanation for your answer.

Your answer

Q9. Let's, $\Sigma = \{a, b\}$. You are given a regular expression, **$a(\Sigma^* a^+ \epsilon) \cup b(\Sigma^* b^+ \epsilon)$** , what is the correct language for this regex?

2 points

Here, $a^+ = aa^*$.

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$.

- a) w starts and ends with the same symbols
- b) w contains either at least two a or at least two b
- c) w contains substring ab
- d) w contains an a followed by a b
- e) w contains equal numbers of ab and ba

Choose the best answer from below.

- ☐ a
- ☐ a and b
- ☐ c and d
- ☐ a and e
- ☐ a, b and e

Write an explanation for your answer.

Your answer _____

Q10. Which of the following Regular expressions are equivalent to **$a(a^*b^* \cup b^*)b$** 2 points

- a) $a^+(ab^+)$
- b) $a^*(ab^+)$
- c) a^+b^+
- d) $a(aa^*b^*)b$
- e) $a^*(a^*b^+ \cup b^*)b$

Choose the best answer from below.

- ☐ a and b
- ☐ c and d
- ☐ d and e
- ☐ b and c

Write an explanation for your answer.

Your answer

Q11. $(01 \cup 1010 \cup 1)^* (00 \cup 010)$

2 points

The length of a string, w can be expressed by $|w|$. Find out how many distinct strings can be generated using the given regular expression such that $3 \leq |w| \leq 5$. Write a numeric value only.

Your answer

Write an explanation for your answer.

Your answer

Q12. You have a regular expression $(0 \cup 1)(0 \cup 1)00^*(0 \cup 1)^*1^*$ And another regular expression $(00 \cup 11 \cup 10 \cup 01)01^*0^*(1 \cup 0 \cup \epsilon)^*0^*$ Are these two the same?

2 points

☐ Yes

☐ No

Write an explanation for your answer.

Your answer

Q13. Let's $\Sigma = \{0,1\}$. What is the regular expression for the strings that have 2 points

even length and start with 1?

- a) $1(0 \cup 1)^*$
- b) $(10 \cup 11)((0 \cup 1)(0 \cup 1))^*$
- c) $(1(0 \cup 1))^*$
- d) $1(0 \cup 1)((0 \cup 1)(0 \cup 1))^*$
- e) $1(0 \cup 1)(00 \cup 01 \cup 11 \cup 10)^* \cup \epsilon$

Select the best answer.

- ☐ d and e
- ☐ b and e
- ☐ b and d
- ☐ b, d and e
- ☐ a, c and e

Write an explanation for your answer.

Your answer

Q14. $L = \{w \in \{0,1\}^*: w \text{ doesn't contain } 001\}$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$.

Which of the following Regular Expressions generate L :

$(1^*(01)^*)^*(\epsilon \cup 0 \cup 000^*)$

1 point

- ☐ Correct
- ☐ Incorrect

☐ Incorrect

$(1^*(0 \cup \epsilon) 1^*)^*$

1 point

☐ Correct

☐ Incorrect

$(1 \cup 10)^* (\epsilon \cup 0 \cup 000^*)$

1 point

☐ Correct

☐ Incorrect

$(1 \cup 01)^* 0^*$

1 point

☐ Correct

☐ Incorrect

$(1^* (10)^*)^* 0^*$

1 point

☐ Correct

☐ Incorrect

Q15. $(0 \cup \epsilon) (10)^* (1 \cup \epsilon)$

Note, for a regular language L , the regular expression will be correct if and only if it generates all the strings, $w \in L$, and doesn't generate any string, $w \notin L$.

Which of the following regular languages are generated by the given regular expression

$L = \{ w \in \{0,1\}^* : w \text{ doesn't contain } 00 \}$

1 point

- ☐ Correct
- ☐ Incorrect

$L = \{ w \in \{0,1\}^* : 0 \text{ and } 1 \text{ alternates in } w \}$

1 point

- ☐ Correct
- ☐ Incorrect

$L = \{ w \in \{0,1\}^* : w \text{ doesn't contain either } 00 \text{ or } 11 \}$

1 point

- ☐ Correct
- ☐ Incorrect

$L = \{ w \in \{0,1\}^* : \text{The period of } w \text{ is either } 01 \text{ or } 10 \}$

1 point

The period of a string is the smallest prefix, when repeated one or more times, can recreate the entire string.

- ☐ Correct
- ☐ Incorrect

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